

DRAWING LIST FOR TEST PACKAGE

GLU-521

PROJECT NO.: 22328D

PROJECT TITLE : M15X Ph2 Project

No	PKG No	UNIT	LINE NUMBER			P&ID		REMARK
			ISO DRAWING No	SHEET	REV	P&ID No	REV	
1	521	4FA	25A-GLU-ST3-N-6514-01	1 OF 3	0	CJ-M15W14-PR-A20-000153	1A	
2	521	4FA	25A-GLU-ST3-N-6514-02	2 OF 3	0	CJ-M15W14-PR-A20-000153	1A	
3	521	4FA	25A-GLU-ST3-N-6514-03	3 OF 3	0	CJ-M15W14-PR-A20-000153	1A	
4	521	4FA	25A-GLU-ST3-N-6515-01	1 OF 3	0	CJ-M15W14-PR-A20-000153	1A	
5	521	4FA	25A-GLU-ST3-N-6515-02	2 OF 3	0	CJ-M15W14-PR-A20-000153	1A	
6	521	4FA	25A-GLU-ST3-N-6515-03	3 OF 3	0	CJ-M15W14-PR-A20-000153	1A	
7	521	4FA	25A-GLU-ST3-N-6516-01	1 OF 1	0	CJ-M15W14-PR-A20-000153	1A	
8	521	4FA	25A-GLU-ST3-N-6517-01	1 OF 2	0	CJ-M15W14-PR-A20-000153	1A	
9	521	4FA	25A-GLU-ST3-N-6517-02	2 OF 2	0	CJ-M15W14-PR-A20-000153	1A	
10	521	4FA	25A-GLU-ST3-N-6518-01	1 OF 2	0	CJ-M15W14-PR-A20-000153	1A	
11	521	4FA	25A-GLU-ST3-N-6518-02	2 OF 2	0	CJ-M15W14-PR-A20-000153	1A	
12	521	4FA	25A-GLU-ST3-N-6519-01	1 OF 1	0	CJ-M15W14-PR-A20-000153	1A	
13	521	4FA	25A-GLU-ST3-N-6520-01	1 OF 1	0	CJ-M15W14-PR-A20-000153	1A	
14	521	4FA	25A-GLU-ST3-N-6521-01	1 OF 1	0	CJ-M15W14-PR-A20-000153	1A	
15	521	4FA	25A-GLU-ST3-N-6522-01	1 OF 1	0	CJ-M15W14-PR-A20-000153	1A	

ITEM NO.	W14-T-GLU-4606 A/B		
SERVICE	GLUCOSE STORAGE TANK		
Q'TY	2		
TYPE	VERTICAL & CYLINDRICAL		
SIZE	ID x H	m	
		3.6 x 4.5 (EH(4.2))	
CAPA.	m3	46 (EH(43))	
MAT'L		STS304	

대외비

KEY PLAN (4F)

- NOTE
- DIAPHRAGM PUMP
MAT'L 표기 : HEAD/DIAPHRAGM/CHECK BALL
작품종류에 따라 LINE 색상을 구분하여야 한다.
 - CHEMICAL UNLOADING STATION에서 LEVEL을 볼 수 있도록 한다.
 - CHEMICAL STORAGE TANK CURB 및 PUMP AREA에서 발생되는 DRAIN은 인근의 TRENCH로 접속되어, DRAIN HEADER로 이송된다.
 - TANK의 역면 레벨 개이치는 VENDOR에서 공급한다.
 - INTERLOCK NO. I-4617
(W14-CDP-GLU-4641A~C, 4642 A~C AUTO STOP)
: REMOTE STATUS에서 W14-T-GLU-4606 A/B LEVEL (LI-W6064 A/B) LOW LOW STOP.
 - INTERLOCK NO. I-4659
(W13-CTP-GLU-4675 AUTO STOP)
: REMOTE STATUS에서 W14-T-GLU-4606 A/B LEVEL (LI-W6063 A/B) HIGH HIGH STOP.
 - PUMP VENDOR 공급.
 - 운전 시 CLOSE 되는 배관에는 LOTO 설치.
 - STRAINER : Y TYPE (6~8 MESH).
 - FIC-W2253, W2263 등 선택운전.
 - OPERATING CAPACITY : 13 LPM
OPERATING PRESSURE : 0.25 MPa

GLU-521

ITEM NO.	W14-CDP-GLU-4641 A/B/C		
SERVICE	CIF MBR GLUCOSE FEED PUMP		
Q'TY	2 + 1		
TYPE	DIAPHRAGM		
CAPA.	l/min.ea x MPa	MAX. 14.9 x MAX 0.3 NOTE 12	
MAT'L		SSC13A/PTFE/STS316	
POWER kW		(0.75)	

ITEM NO.	W14-CDP-GLU-4642 A/B/C		
SERVICE	CIF MBR 2'nd GLUCOSE FEED PUMP		
Q'TY	2 + 1		
TYPE	DIAPHRAGM		
CAPA.	l/min.ea x MPa	MAX. 14.9 x MAX 0.3 NOTE 12	
MAT'L		SSC13A/PTFE/STS316	
POWER kW		(0.75)	

2A	REVISED AFC	25.08.14
2	REVISED AFC	25.01.17
1	REVISED AFC	24.10.31
수정	실명	날짜
REV.	DESCRIPTION	DATE
제	도	이 인 학
검	토	한 진 회
인	PROVED BY	박 동 명
제	작	사

SK 에코플랜트

사명
PROJECT TITLE

M15X Ph4 WWT Project

도면명
DWG. TITLE

PIPING & INSTRUMENT DIAGRAM
GLUCOSE STORAGE TANK
& FEED PUMP

속척
SCALE

NONE

날짜
DATE

2025. 08. 14

PROJECT NO.

22330D

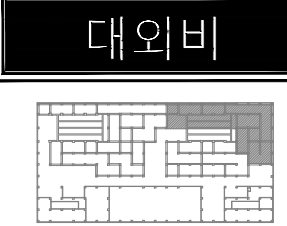
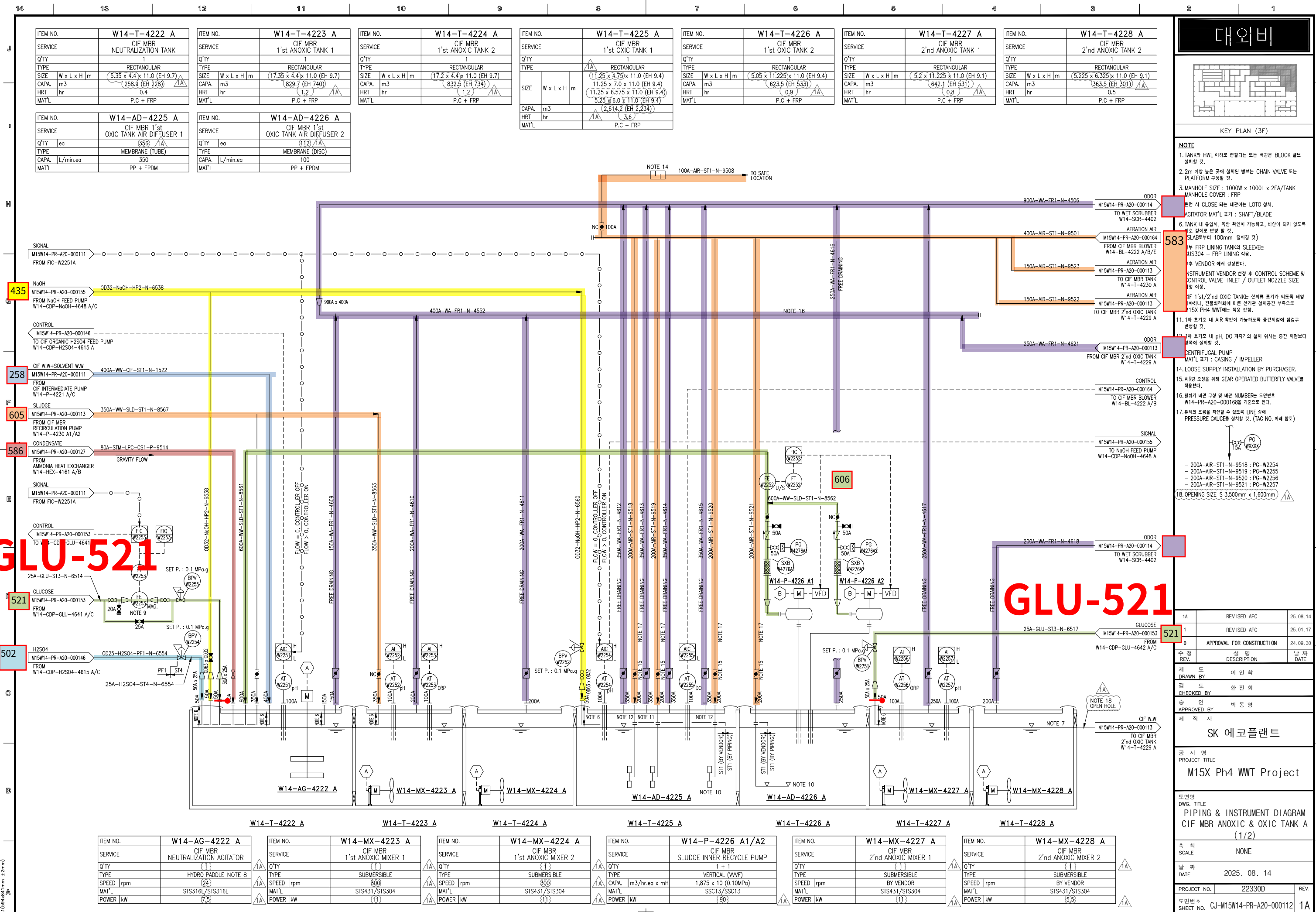
REV.

A

도면번호
SHEET NO.

CJ-M15W14-PR-A20-000153

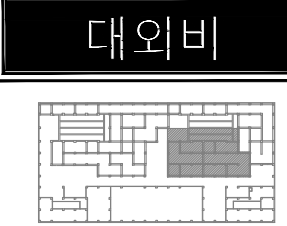
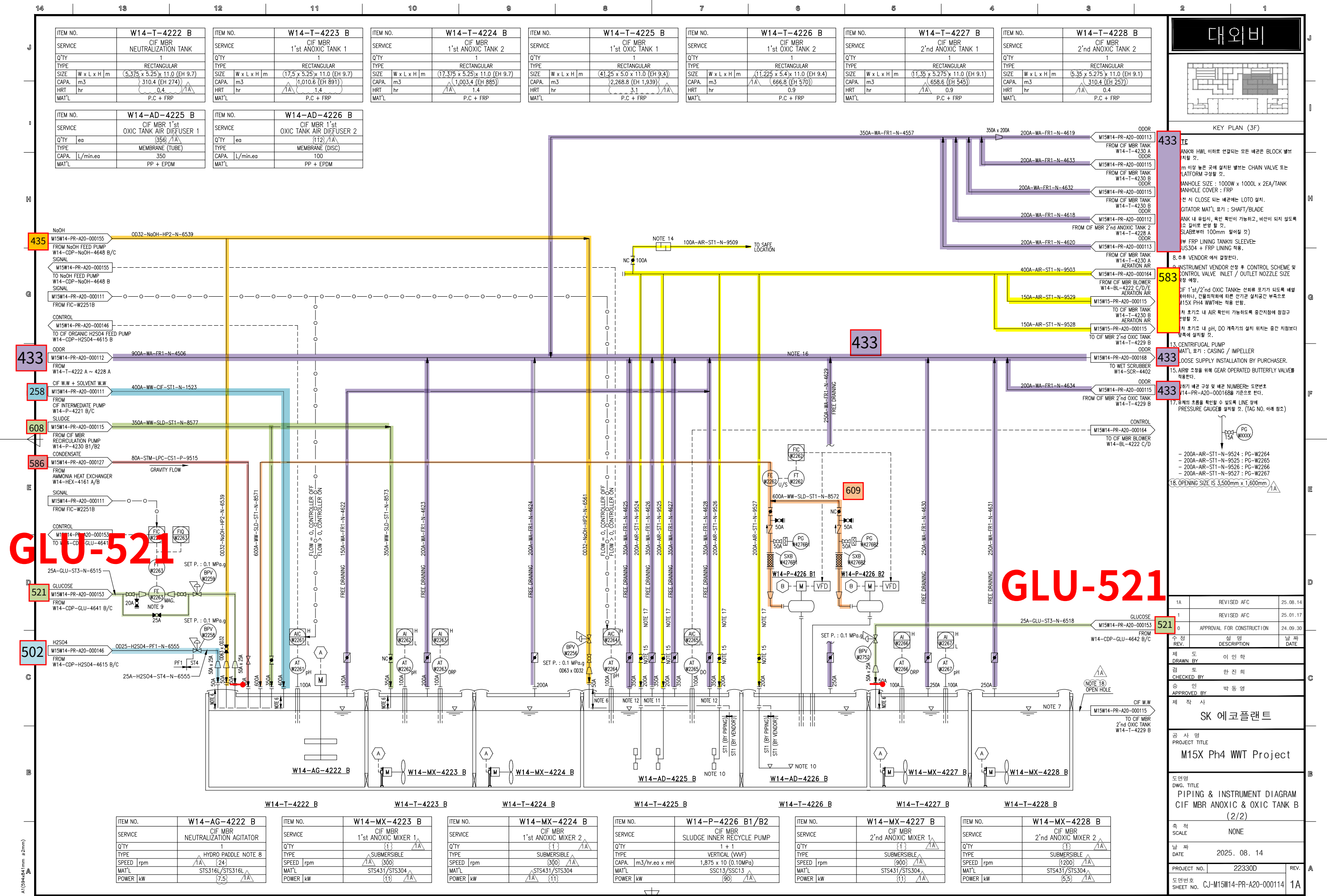
2A



KEY PLAN (3F)

- NOTE**
1. TANK의 HWL 이하로 연결되는 모든 배관은 BLOCK 밸브 설치할 것.
 2. 2m 이상 높은 곳에 설치된 밸브는 CHAIN VALVE 또는 PLATFORM 구성할 것.
 3. MANHOLE SIZE : 1000W x 1000L x 2EA/TANK MANHOLE COVER : FRP
 4. 운전 시 CLOSE 되는 배관에는 LOTO 설치.
 5. AGITATOR MAT'L 표기 : SHAFT/BLADE
 6. TANK 내 유입시, 육안 확인이 가능하고, 배선이 되지 않도록 최소 길이로 반영 할 것.
 7. FRP LINING TANK의 SLEEVE는 SUS304 + FRP LINING 적용.
 8. 이후 VENDOR 에서 결정한다.
 9. INSTRUMENT VENDOR 선정 후 CONTROL SCHEME 및 CONTROL VALVE INLET / OUTLET NOZZLE SIZE 결정 예정.
 10. CIF 1st/2nd OXIC TANK는 전회류 포기가 되도록 배설해야하나, 건물외곽에 따른 산기온 설치공간 부족으로 M15X Ph4 WWT에는 적용 안함.
 11. 1차 호기조 내 AIR 확인이 가능하도록 중간지점에 점검구 반영할 것.
 12. 1차 호기조 내 pH, DO 계측기의 설치 위치는 중간 지점보다 상부에 설치할 것.
 13. CENTRIFUGAL PUMP MAT'L 표기 : CASING / IMPELLER
 14. LOOSE SUPPLY INSTALLATION BY PURCHASER.
 15. AIR용 조정을 위해 GEAR OPERATED BUTTERFLY VALVE를 적용한다.
 16. 말하기 배관 구성 및 배관 NUMBER는 도면번호 W14-PR-A20-000168을 기준으로 한다.
 17. 유체의 흐름을 확인할 수 있도록 LINE 상에 PRESSURE GAUGE를 설치할 것. (TAG NO. 아래 참조)
 18. OPENING SIZE IS 3,500mm x 1,600mm

- 1A. REVISED AFC 25.08.14
1. REVISED AFC 25.01.17
0. APPROVAL FOR CONSTRUCTION 24.09.30
- 수정 REV. 설명 DESCRIPTION 날짜 DATE
- 제 도 DRAWN BY 이 인 학
- 검 토 CHECKED BY 한 진 희
- 승 인 APPROVED BY 박 동 영
- 제 작 사 SK 에코플랜트
- 공 사 명 PROJECT TITLE M15X Ph4 WWT Project
- 도면명 DWG. TITLE PIPING & INSTRUMENT DIAGRAM CIF MBR ANOXIC & OXIC TANK A (1/2)
- 속 척 SCALE NONE
- 날 짜 DATE 2025. 08. 14
- PROJECT NO. 22330D REV. A
- 도면번호 SHEET NO. CJ-M15W14-PR-A20-000112 R1 P&ID FOR CIF M



- 433. ANK의 HWL 이하로 연결되는 모든 배관은 BLOCK 별반 치할 것.
- m 이상 높은 곳에 설치된 별반은 CHAIN VALVE 또는 LATIFORM 구성할 것.
- ANHOLE SIZE : 1000W x 1000L x 2EA/TANK ANHOLE COVER : FRP
- 전 시 CLOSE 되는 배관에는 LOTO 설치.
- GITATOR MAT'L 표기 : SHAFT/BLADE
- ANK 내 유입시, 육안 확인이 가능하고, 비산이 되지 않도록 소 길이로 변경 할 것.
- SLAB로부터 100mm 떨어진 것
- FRP LINING TANK의 SLEEVE는 US304 + FRP LINING 적용.
- 8. 주후 VENDOR에서 결정한다.
- 9. INSTRUMENT VENDOR 선정 후 CONTROL SCHEME 및 CONTROL VALVE INLET / OUTLET NOZZLE SIZE 결정 예정.
- 10. C/F 1st/2nd OXIC TANK는 선회류 포기가 되도록 배설해야하나, 건물외벽에 따른 순간 설치공간 부족으로 15X PH4 WWT에는 적용 안함.
- 자 호기조 내 AIR 확인이 가능하도록 중간지점에 잠금구 구성할 것.
- 자 호기조 내 pH, DO 계측기의 설치 위치는 중간 지점보다 양쪽에 설치할 것.
- 13. CENTRIFUGAL PUMP MAT'L 표기 : CASING / IMPELLER
- DOOSE SUPPLY INSTALLATION BY PURCHASER.
- 15. AIR형 조정을 위해 GEAR OPERATED BUTTERFLY VALVE를 적용한다.
- 433. 배기 배관 구성 및 배관 NUMBER는 도면번호 W14-PR-A20-000168을 기준으로 한다.
- 17. 유체의 흐름을 확인할 수 있도록 LINE 상에 PRESSURE GAUGE를 설치할 것. (TAG NO. 아래 참조)

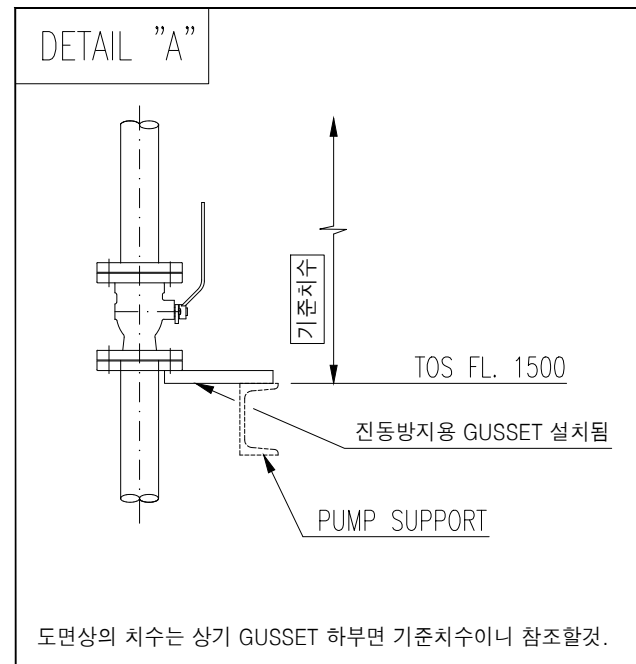
- 200A-AIR-ST1-N-9524 : PG-W2264
- 200A-AIR-ST1-N-9525 : PG-W2265
- 200A-AIR-ST1-N-9526 : PG-W2266
- 200A-AIR-ST1-N-9527 : PG-W2267

18. OPENING SIZE IS 3,500mm x 1,600mm

1A	REVISED AFC	25.08.14
1	REVISED AFC	25.01.17
0	APPROVAL FOR CONSTRUCTION	24.09.30

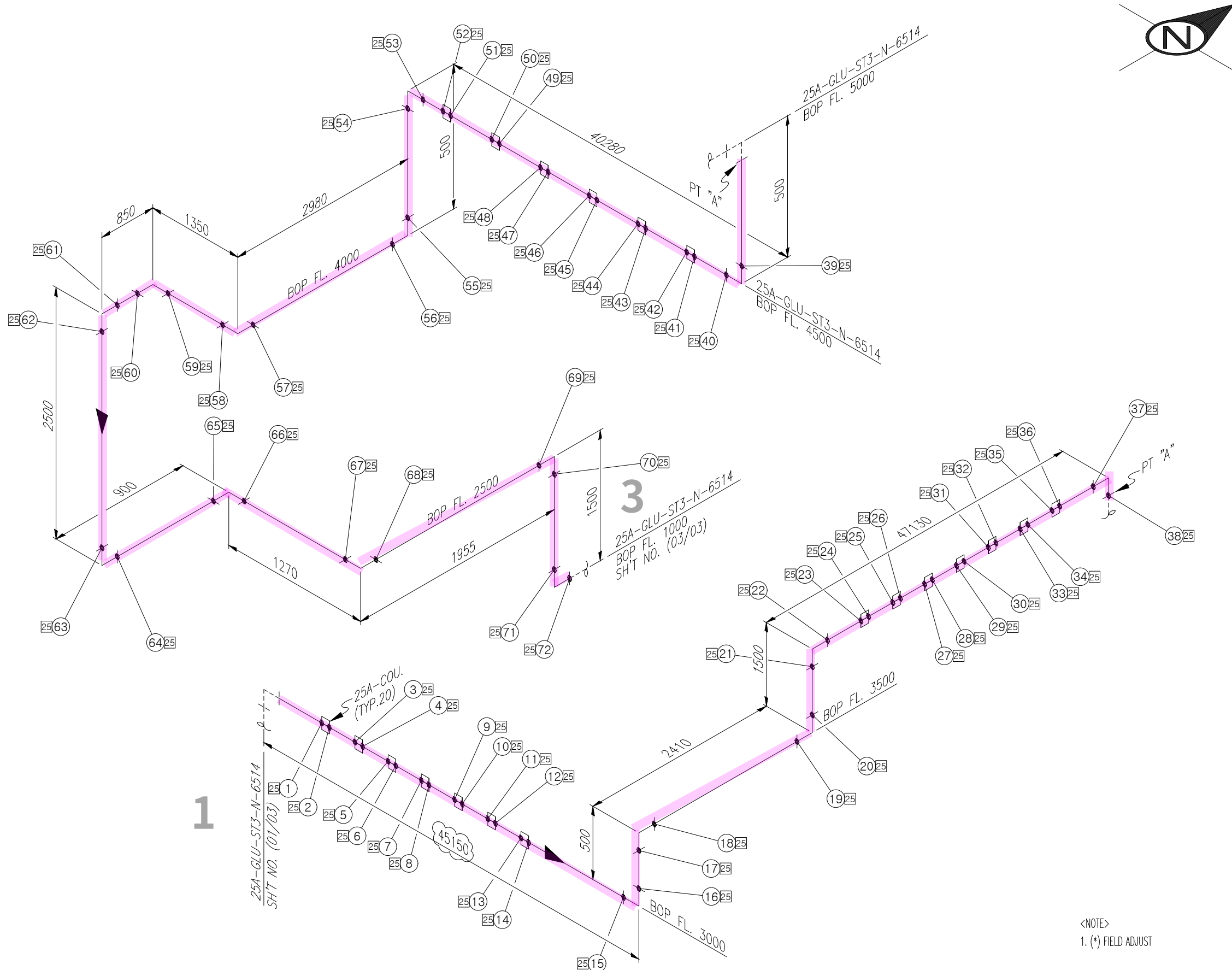
수정	실명	날짜
REV.	DESCRIPTION	DATE
제	도	이인학
DRAWN BY		
검	토	한진희
CHECKED BY		
승	인	박동영
APPROVED BY		
제	작	사

SK 에코플랜트		
공 사 명		
PROJECT TITLE		
M15X Ph4 WWT Project		
도면명		
DWG. TITLE		
PIPING & INSTRUMENT DIAGRAM		
CIF MBR ANOXIC & OXIC TANK B		
(2/2)		
속 척		
SCALE		
NONE		
날 짜		
DATE		
2025. 08. 14		
PROJECT NO.		22330D
도면번호		CJ-M15W14-PR-A20-000114
SHEET NO.		1A

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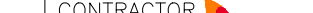
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INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA W14							CONTRACTOR	PROJECT NAME		
	THICK.	—	DESIGN	3.4	60	UNIT 4FA								M15X Ph4 Project		
			OPERATING	3.0	AMB	REF. PLAN DWG. CJ-M15W14-PI-A30-040012-A							COMMODITY	GLU	JOB NO.	
								25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영			22330D	
						REF. P & ID CJ-M15W14-PR-A20-000153	REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D	LINE NO.	25A-GLU-ST3-N-6514	SH'T NO. 01 / 03	REV. 0



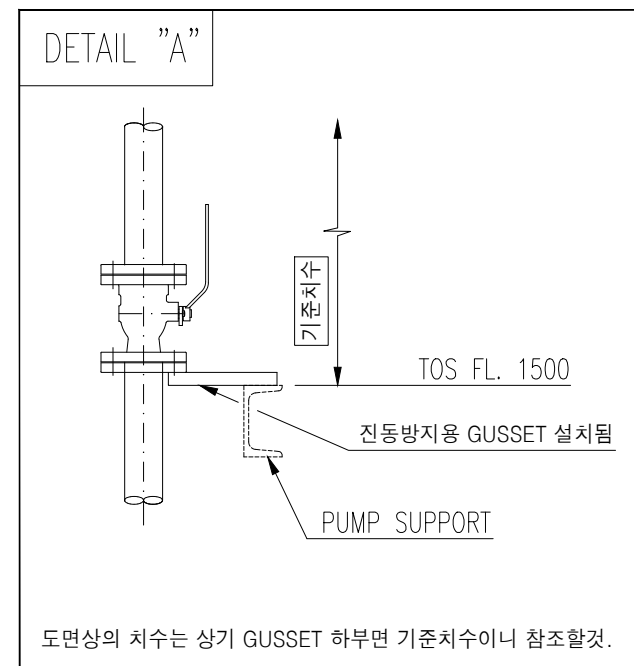
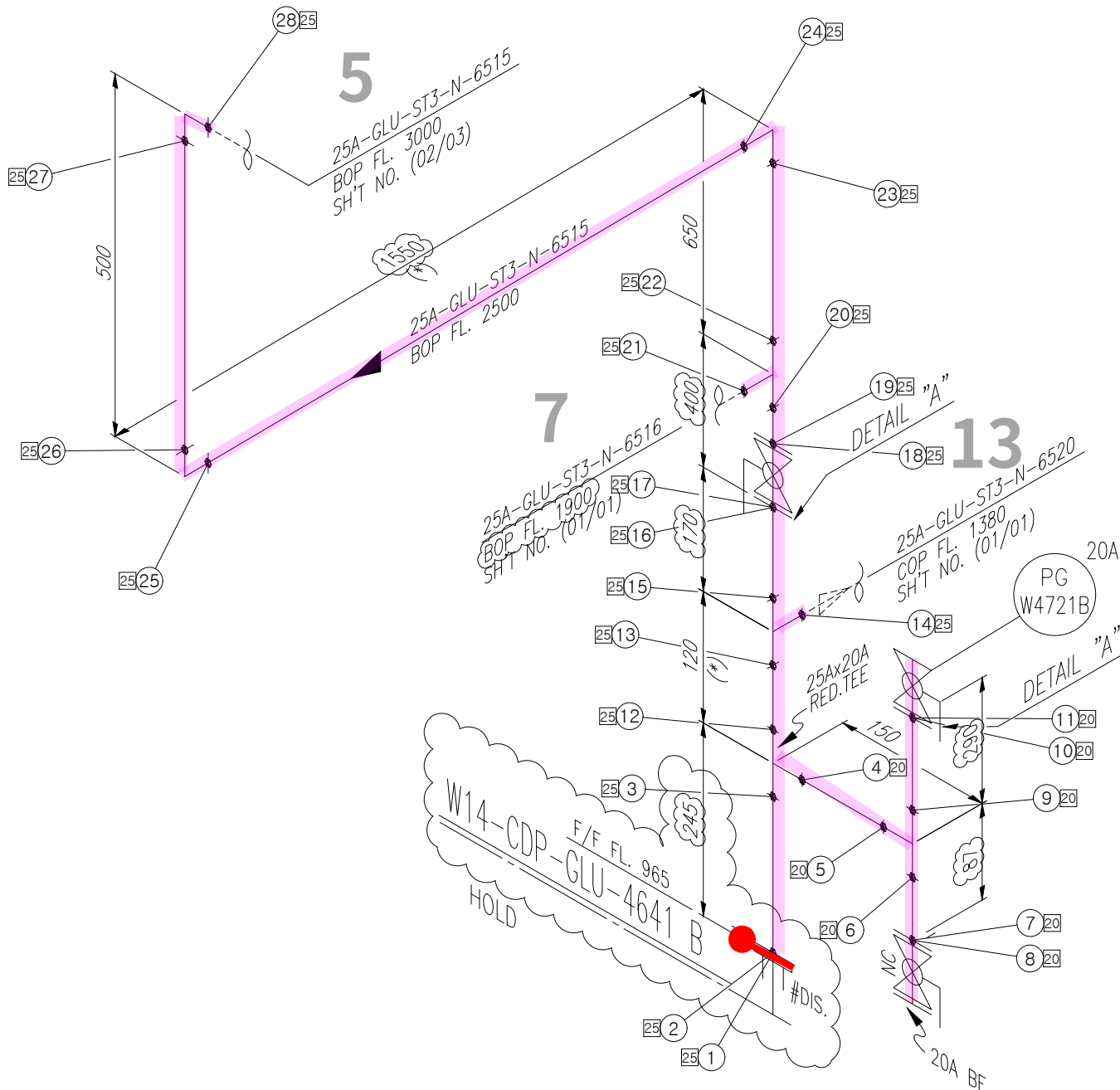
<NOTE>
1. (*) FIELD ADJUST

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VB1/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14								CONTRACTOR	PROJECT NAME	
	THICK.	—	DESIGN	3.4	60	UNIT	4FA								M15X Ph4 Project		
			OPERATING	3.0	AMB	REF. PLAN DWG. CJ-M15W14-PI-A30-040009-A								COMMODITY	GLU	JOB NO.	
						REF. P & ID CJ-M15W14-PR-A20-000153			25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영	LINE NO.	25A-GLU-ST3-N-6514	SH'T NO. 02 / 03	REV. 0
								REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D				



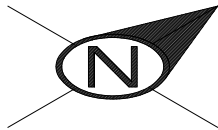
ITEM	SIZE	Q'TY	DESCRIPTION
ST3			
P	25A	3.6M	PIPE, STS 304TP, ERW, PE – SCH 10S
P	20A	0.6M	PIPE, STS 304TP, ERW, PE – SCH 10S
E90	25A	3EA	ELBOW 90 DEG, STSF304, SW – SCH 10S
T	25A	2EA	TEE EQUAL, STSF304, SW – SCH 10S
T	20A	1EA	TEE EQUAL, STSF304, SW – SCH 10S
TR	25Ax20A	1EA	TEE RED, STSF304, SW – SCH 10S
VB	25A	1EA	BALL VALVE, KS 10K, RF FLG'D, BODY–STSF304 /
			TRIM–ST304 / SEAT–PTFE, FB, LEVER
VB	20A	2EA	BALL VALVE, KS 10K, RF FLG'D, BODY–STSF304 /
			TRIM–ST304 / SEAT–PTFE, FB, LEVER
F	25A	3EA	FLG SO, KS 10K, RF, STS304
F	20A	2EA	FLG SO, KS 10K, RF, STS304
FB	20A	1EA	FLG BLIND, KS 10K, RF, STS304
G	25A	3EA	GASKET, KS 10K, RF, PTFE+EPDM, 3.0MM TH'K
G	20A	3EA	GASKET, KS 10K, RF, PTFE+EPDM, 3.0MM TH'K
B	25A	3SET	MACHINE BOLT & NUTS :STS 304 / W WASHERS
B	20A	3SET	MACHINE BOLT & NUTS :STS 304 / W WASHERS
PG	20A	1EA	PRESSURE GAUGE, KS 10K, RF
LOTO	20A	1EA	LOTO



- <NOTE>
1. TRAY & ANGLE, CHANNEL SUPPORT가 적용되는
배관 ELEVATION은 현장 맞춤으로 한다.
2. (*) FIELD ADJUST

* PIPE : P/P1	* 90° ELBOW : E90	* 45° ELBOW : E45	* TEE : T	* RED.TEE : TR	* STUB-IN : STU	* REINFORCING PAD : RP	* CON.RED : RC	* ECC.RED : RE	* CON.SWAGE : RCS/RCS1	* ECC.SWAGE : RES/RES1	* NIPPLE : NP/NP1/NP2	* CAP : CAP/CAP1	* PLUG : PLG	* FULL COUPLING : FC	* HALF COUPLING : HF/HF1
* STUB-IN : STU	* SOCKET(HDPE/PVC) : SC	* SADDLE(HDPE) : SD	* FLANGE : F/F1	* GASKET : G/G1/G2	* BOLT & NUT : B/B1	* BLIND FLANGE : FB/FB1	* GATE VA. : VG/VG1/VG2	* BALL VA. : VB/VB0/VB1/VB10/VB2/VB20	* BUTTERFLY VA. : VF/VF4	* CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7					
* GLOBE VA. : VL/VL1/VL2	* DIAPHRAGM VA. : VD/VD0	* DAMPER : VP	* QUICK COUPLING : SQC	* FLEXIBLE JOINT(STEEL) : FLXB	* FLEXIBLE JOINT(RUBBER) : FLXR	* FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL	* FLEXIBLE HOSE : FLXH	* STEAM TRAP : ST	* SPRAY NOZZLE : SPN	* EMERGENCY SHOWER : SSH					
* STRAINER(Y-TYPE) : YSTR	* STRAINER(T-TYPE) : TSTR	* STRAINER(BUCKET-TYPE) : BSTR	* ANCHOR PLATE : STOP	* BIRD SCREEN : BS	* ISOLATION KIT : IK	* UNLOADING BOX : UB	* AIR TRAP : AT	* LOTO : LOTO	* SIGHT GLASS : SG	* PRESSURE GAUGE : PG	* SHILED : SH	* SPECIAL BOLT & NUT : BL/BL1			

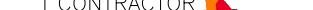
INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA W14								CONTRACTOR	PROJECT NAME	
	THICK.	—	DESIGN	3.4	60	UNIT 4FA									M15X Ph4 Project	
			OPERATING	3.0	AMB	REF. PLAN DWG. CJ-M15W14-PI-A30-040012-A		25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영	COMMODITY	GLU	JOB NO.	
						REF. P & ID CJ-M15W14-PR-A20-000153							REV. NO.	DATE	DESCRIPTION	BY

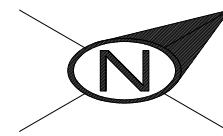


CLIENT



5

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14							CONTRACTOR	PROJECT NAME	
	THICK.	—	DESIGN	3.4	60	UNIT	4FA								M15X Ph4 Project	
			OPERATING	3.0	AMB	REF. PLAN DWG.								COMMODITY	JOB NO.	
						CJ-M15W14-PI-A30-040009-A			25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영	GLU	22330D	
						REF. P & ID		REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D	LINE NO.	SH'T NO.	REV.
						CJ-M15W14-PR-A20-000153								25A-GLU-ST3-N-6515	02 / 03	0

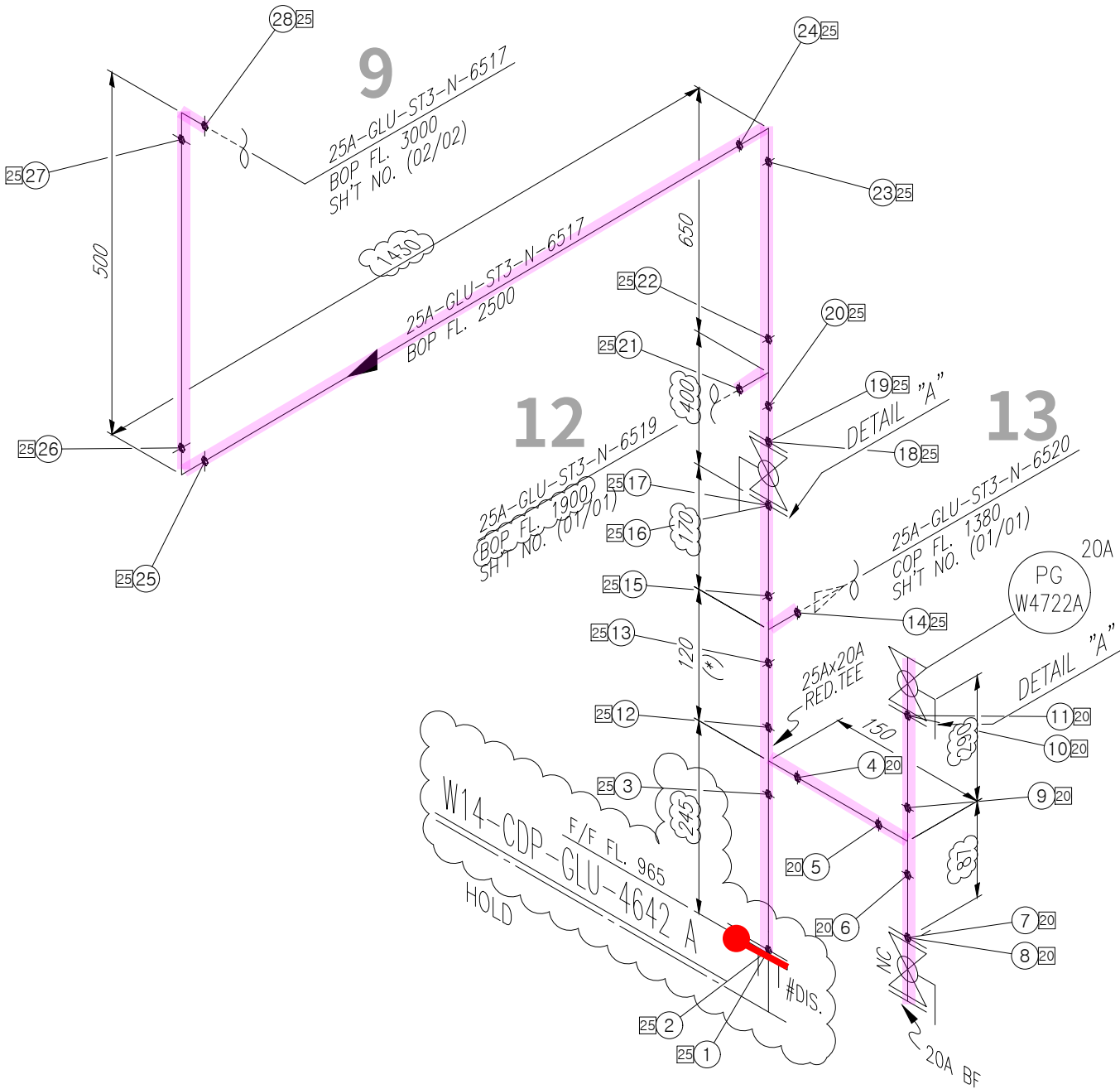
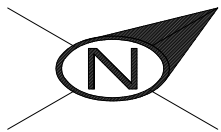


CLIENT

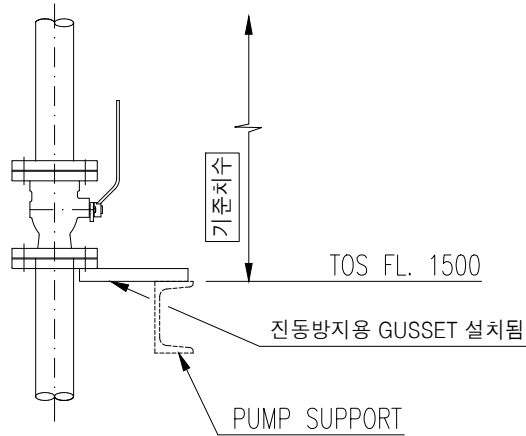


7

또한, SK HYNIX의 기업비밀 자료로서, 어떠한 경우에도 제3자에게 인계, 누설, 제공, 제3자에게 양도할 수 없으며, 회사나 유무포자의 사용에 의해서 발생하는 유무포자에 대한 손해배상책임을 면할 수 있는



DETAIL "A"



도면상의 치수는 상기 GUSSET 하부면 기준치수이니 참조할것.

- <NOTE>
1. TRAY & ANGLE, CHANNEL SUPPORT가 적용되는 배관 ELEVATION은 현장 맞춤으로 한다.
2. (*) FIELD ADJUST



ITEM	SIZE	Q'TY	DESCRIPTION
ST3			
P	25A	3.5M	PIPE, STS 304TP, ERW, PE - SCH 10S
P	20A	0.6M	PIPE, STS 304TP, ERW, PE - SCH 10S
E90	25A	3EA	ELBOW 90 DEG, STSF304, SW - SCH 10S
T	25A	2EA	TEE EQUAL, STSF304, SW - SCH 10S
T	20A	1EA	TEE EQUAL, STSF304, SW - SCH 10S
TR	25Ax20A	1EA	TEE RED, STSF304, SW - SCH 10S
VB	25A	1EA	BALL VALVE, KS 10K, RF FLG'D, BODY-STSF304 / TRIM-STSF304 / SEAT-PTFE, FB, LEVER
VB	20A	2EA	BALL VALVE, KS 10K, RF FLG'D, BODY-STSF304 / TRIM-STSF304 / SEAT-PTFE, FB, LEVER
F	25A	3EA	FLG SO, KS 10K, RF, STS304
F	20A	2EA	FLG SO, KS 10K, RF, STS304
FB	20A	1EA	FLG BLIND, KS 10K, RF, STS304
G	25A	3EA	GASKET, KS 10K, RF, PTFE+EPDM, 3.0MM TH'K
G	20A	3EA	GASKET, KS 10K, RF, PTFE+EPDM, 3.0MM TH'K
B	25A	3SET	MACHINE BOLT & NUTS :STS 304 / W WASHERS
B	20A	3SET	MACHINE BOLT & NUTS :STS 304 / W WASHERS
PG	20A	1EA	PRESSURE GAUGE, KS 10K, RF
LOTO	20A	1EA	LOTO

CLIENT

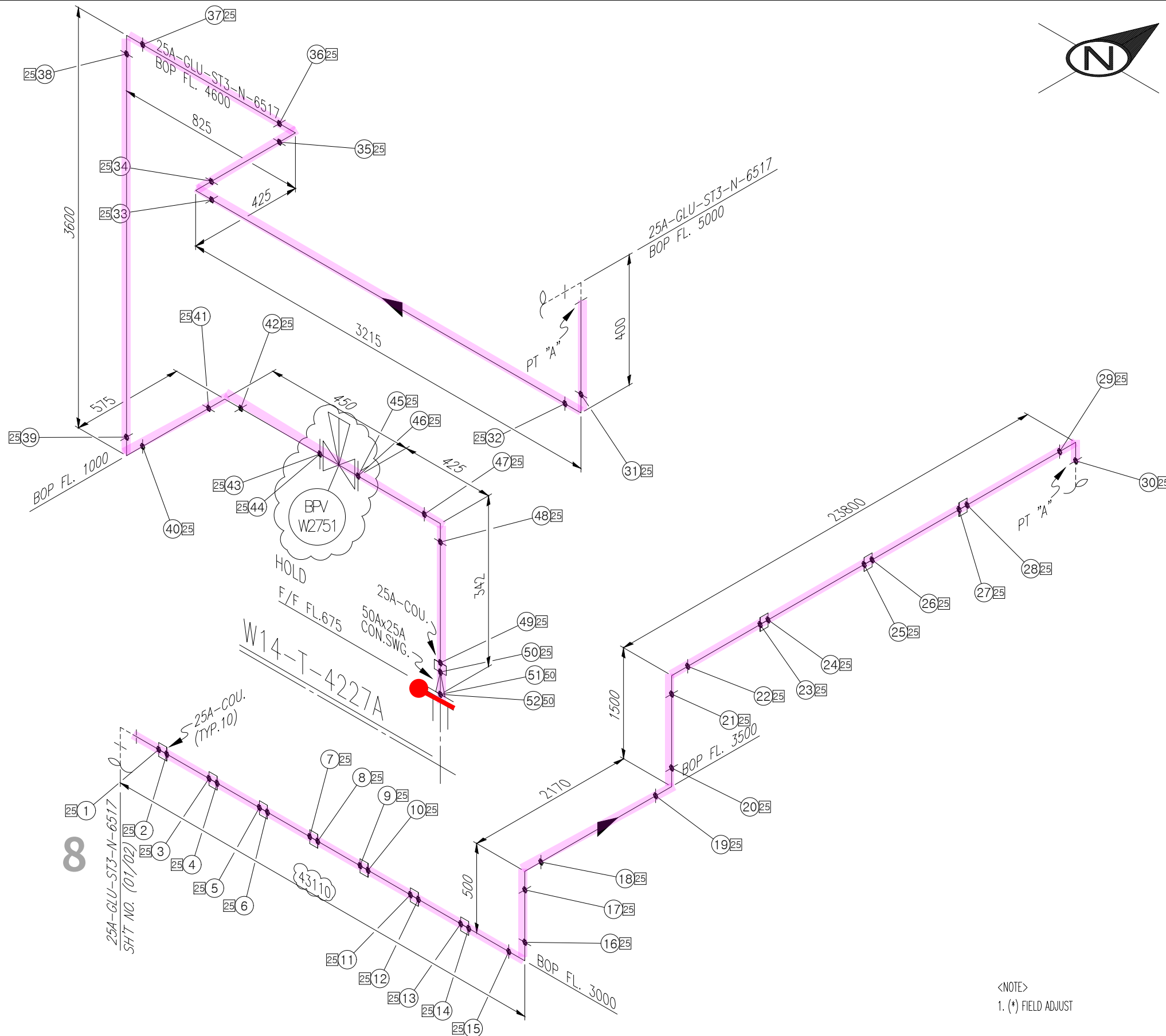


8

CONTRACTOR	PROJECT NAME
SK ecoplant	M15X Ph4 Project
COMMODITY	JOB NO.
GLU	22330D
LINE NO.	SH'T NO. REV.
25A-GLU-ST3-N-6517	01 / 02 0

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

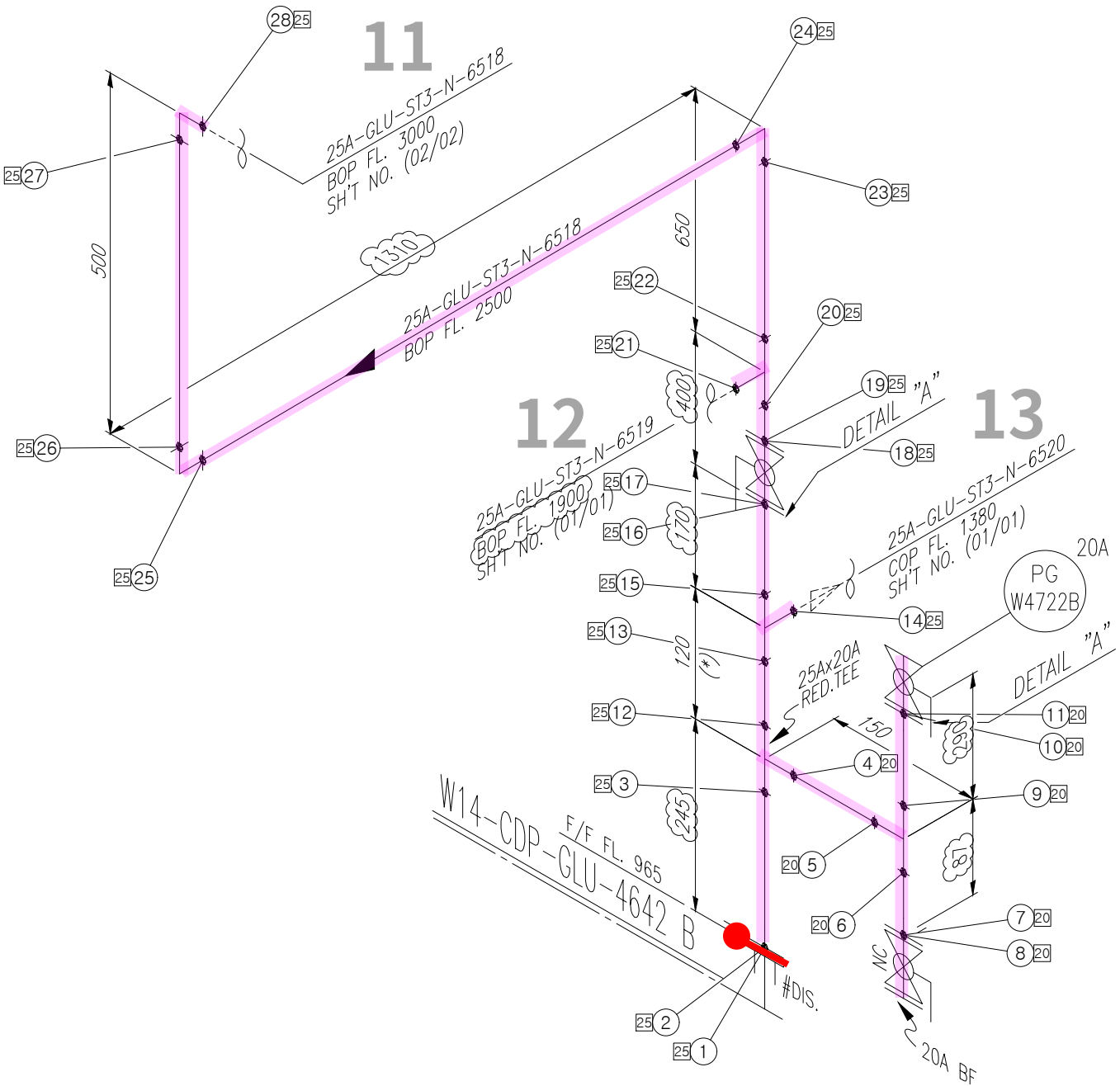
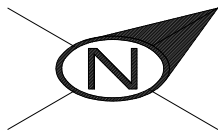
INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14	REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D	CONTRACTOR	PROJECT NAME
	THICK.	-	DESIGN	3.4	60	UNIT	4FA								
			OPERATING	3.0	AMB	REF. PLAN DWG.									
						CJ-M15W14-PI-A30-040012-A									
						REF. P & ID									
						CJ-M15W14-PR-A20-000153									



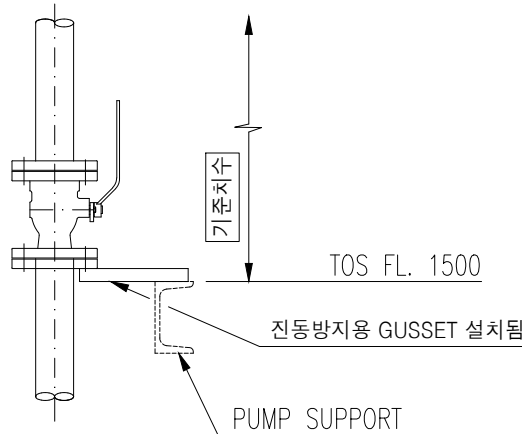
SK ecoplant			
ITEM	SIZE	Q'TY	DESCRIPTION
ST3			
P	25A	81.4M	PIPE, STS 304TP, ERW, PE – SCH 10S
E90	25A	12EA	ELBOW 90 DEG, STSF304, SW – SCH 10S
RCS	50Ax25A	1EA	SWAGE CON, STS304, ERW, BLE/PSE – SCH 10S
FC	25A	11EA	COUPLING FULL, STSF304, SW – SCH 10S
F1	50A	1EA	FLG SO, KS 10K, RF, STS304
F	25A	2EA	FLG SO, KS 10K, RF, STS304
G1	50A	1EA	GASKET, KS 10K, RF, PTFE+EPDM, 3.0MM TH'K
G	25A	2EA	GASKET, KS 10K, RF, PTFE+EPDM, 3.0MM TH'K
B	50A	1SET	MACHINE BOLT & NUTS :STS 304 / W WASHERS
B	25A	2SET	MACHINE BOLT & NUTS :STS 304 / W WASHERS
BPV	25A	1EA	BACK PRESSURE VALVE

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14	0	25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영	CONTRACTOR	PROJECT NAME
	THICK.	-	DESIGN	3.4	60	UNIT	4FA							SK ecoplant	M15X Ph4 Project
			OPERATING	3.0	AMB	REF. PLAN DWG.	CJ-M15W14-PI-A30-040009-A							COMMODITY	JOB NO.
						REF. P & ID	CJ-M15W14-PR-A20-000153	△	REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D	22330D
														LINE NO.	SH'T NO.
														25A-GLU-ST3-N-6517	02 / 02
															REV. 0



DETAIL "A"



도면상의 치수는 상기 GUSSET 하부면 기준치수이니 참조할것.

<NOTE>

1. TRAY & ANGLE, CHANNEL SUPPORT가 적용되는 배관 ELEVATION은 현장 맞춤으로 한다.
2. (*) FIELD ADJUST



ITEM	SIZE	Q'TY	DESCRIPTION
ST3			
P	25A	3.4M	PIPE, STS 304TP, ERW, PE - SCH 10S
P	20A	0.6M	PIPE, STS 304TP, ERW, PE - SCH 10S
E90	25A	3EA	ELBOW 90 DEG, STSF304, SW - SCH 10S
T	25A	2EA	TEE EQUAL, STSF304, SW - SCH 10S
T	20A	1EA	TEE EQUAL, STSF304, SW - SCH 10S
TR	25Ax20A	1EA	TEE RED, STSF304, SW - SCH 10S
VB	25A	1EA	BALL VALVE, KS 10K, RF FLG'D, BODY-STSF304 / TRIM-STSF304 / SEAT-PTFE, FB, LEVER
VB	20A	2EA	BALL VALVE, KS 10K, RF FLG'D, BODY-STSF304 / TRIM-STSF304 / SEAT-PTFE, FB, LEVER
F	25A	3EA	FLG SO, KS 10K, RF, STS304
F	20A	2EA	FLG SO, KS 10K, RF, STS304
FB	20A	1EA	FLG BLIND, KS 10K, RF, STS304
G	25A	3EA	GASKET, KS 10K, RF, PTFE+EPDM, 3.0MM TH'K
G	20A	3EA	GASKET, KS 10K, RF, PTFE+EPDM, 3.0MM TH'K
B	25A	3SET	MACHINE BOLT & NUTS :STS 304 / W WASHERS
B	20A	3SET	MACHINE BOLT & NUTS :STS 304 / W WASHERS
PG	20A	1EA	PRESSURE GAUGE, KS 10K, RF
LOTO	20A	1EA	LOTO

CLIENT

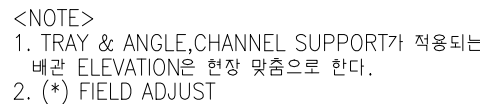
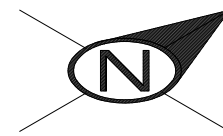


10

CONTRACTOR	PROJECT NAME
SK ecoplant	M15X Ph4 Project
COMMODITY	JOB NO.
GLU	22330D
LINE NO.	SH'T NO. REV.
25A-GLU-ST3-N-6518	01 / 02 0

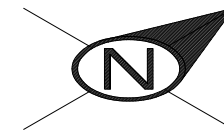
* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14	REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D	CONTRACTOR	PROJECT NAME
	THICK.	-	DESIGN	3.4	60	UNIT	4FA								
			OPERATING	3.0	AMB	REF. PLAN DWG.									
						CJ-M15W14-PI-A30-040012-A									
						REF. P & ID									
						CJ-M15W14-PR-A20-000153									



CLIENT  12

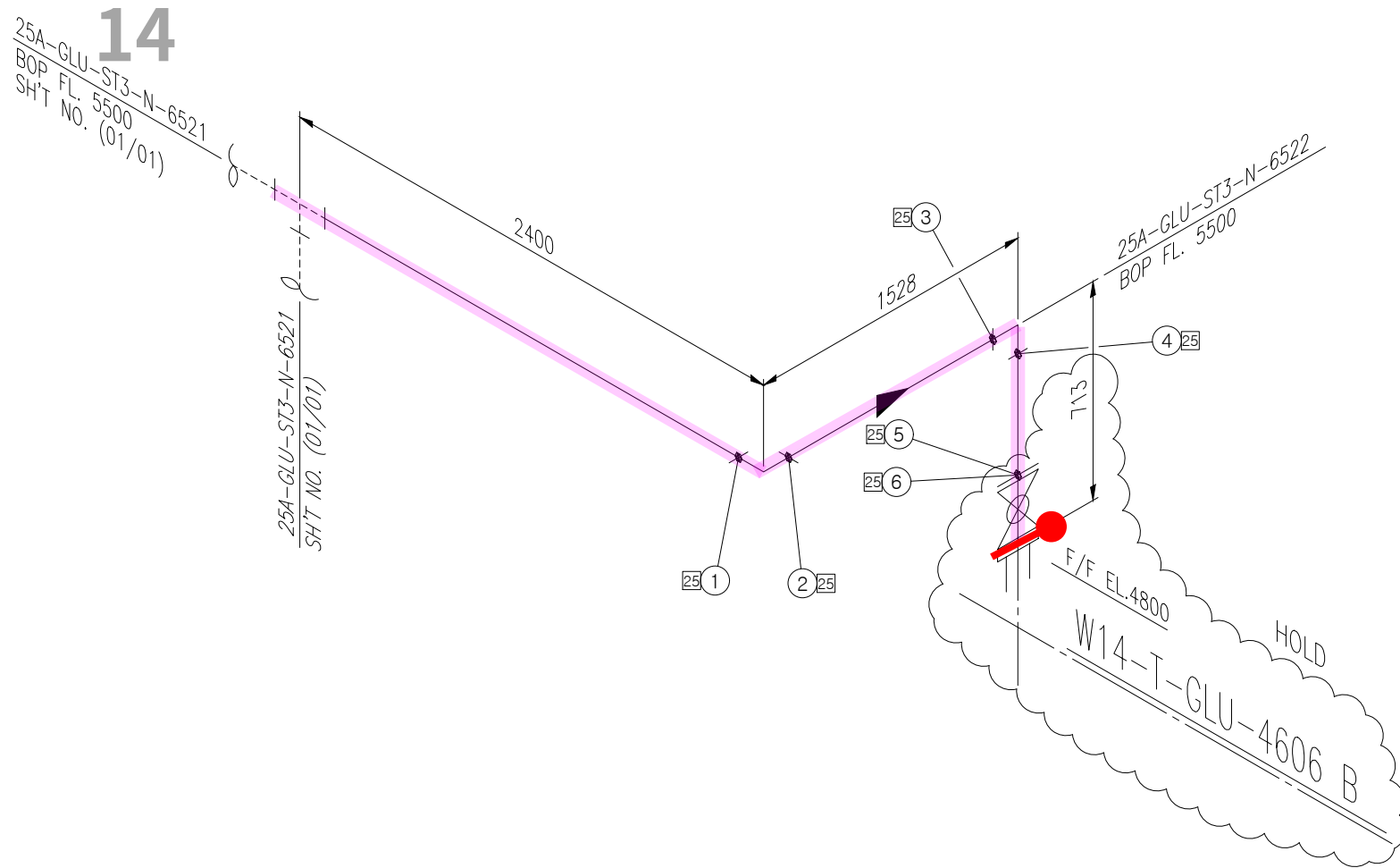
경고 : 본 문건은 SK HYNIX의 기밀비밀 자료로서, 어떠한 경우에도 무단 복제·배포, 제3자에게 공개, 누설 또는 기타 방법으로 유출하여서는 안되며, 회사는 은포자의 귀책사유로 인해 피해가 발생할 경우 은포자에게 민/형사상 책임을 물을 수 있습니다.



1. (*) FIELD ADJUST

CLIENT  14

경고 : 본 도면은 SK HYNIX의 기업비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음

[illegible]

<NOTE>

1. (*) FIELD ADJUST

* PIPE : P/P1	* 90° ELBOW : E90	* 45° ELBOW : E45	* TEE : T	* RED.TEE : TR	* STUB-IN : STU	* REINFORCING PAD : RP	* CON.RED : RC	* ECC.RED : RE	* CON.SWAGE : RCS/RCS1	* ECC.SWAGE : RES/RES1	* NIPPLE : NP/NP1/NP2	* CAP : CAP/CAP1	* PLUG : PLG	* FULL COUPLING : FC	* HALF COUPLING : HF/HF1
* STUB-IN : STU	* SOCKET(HDPE/PVC) : SC	* SADDLE(HDPE) : SD	* FLANGE : F/F1	* GASKET : G/G1/G2	* BOLT & NUT : B/B1	* BLIND FLANGE : FB/FB1	* GATE VA. : VG/VG1/VG2	* BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO	* BUTTERFLY VA. : VF/VF4	* CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7					
* GLOBE VA. : VL/VL1/VL2	* DIAPHRAGM VA. : VD/VDLO	* DAMPER : VP	* QUICK COUPLING : SQC	* FLEXIBLE JOINT(STEEL) : FLXB	* FLEXIBLE JOINT(RUBBER) : FLXR	* FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL	* FLEXIBLE HOSE : FLXH	* STEAM TRAP : ST	* SPRAY NOZZLE : SPN	* EMERGENCY SHOWER : SSH					
* STRAINER(Y-TYPE) : YSTR	* STRAINER(T-TYPE) : TSTR	* STRAINER(BUCKET-TYPE) : BSTR	* ANCHOR PLATE : STOP	* BIRD SCREEN : BS	* ISOLATION KIT : IK	* UNLOADING BOX : UB	* AIR TRAP : AT	* LOTO : LOTO	* SIGHT GLASS : SG	* PRESSURE GAUGE : PG	* SHILED : SH	* SPECIAL BOLT & NUT : BL/BL1			

CLIENT



15

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA W14							CONTRACTOR	PROJECT NAME	
	THICK.	—	DESIGN	1	60	UNIT 4FA								M15X Ph4 Project	
			OPERATING	ATM	AMB	REF. PLAN DWG. CJ-M15W14-PI-A30-040004-D		25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영	COMMODITY	GLU	JOB NO.
					REF. P & ID CJ-M15W14-PR-A20-000153										
						REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D	LINE NO.	25A-GLU-ST3-N-6522	SH'T NO. 01 / 01	REV. 0

경고 : 본 도면은 SK HYNIX의 기밀비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음